

Plane Geometry

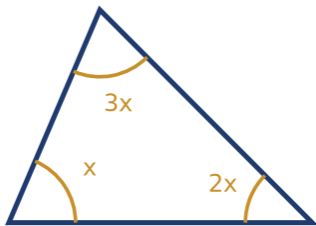


Angle relationships

- 1 Two angles are supplementary, and one is 3 times the other. Find both angles.
- 2 Two parallel lines are cut by a transversal, creating an angle of 70° . Find the measure of its alternate interior angle and its corresponding angle.

Triangle angle sum and exterior angles

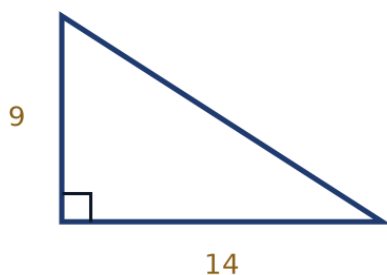
- 3 A triangle has angles measuring x , $2x$, and $3x$. Find each angle.



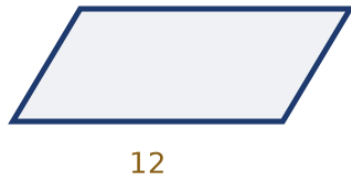
- 4 A triangle's exterior angle at one vertex measures 115° . If one of the remote interior angles is 50° , find the other remote interior angle.

Area and perimeter of polygons

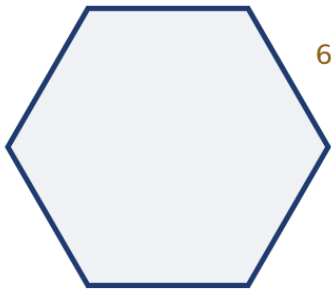
- 5 Find the area of a triangle with base 14 and height 9.



- 6 Find the area of a parallelogram with base 12 and height 5.

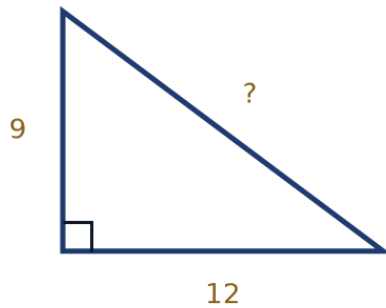


- 7 Find the area of a regular hexagon with side length 6, using the formula $A = \frac{3\sqrt{3}}{2}s^2$.

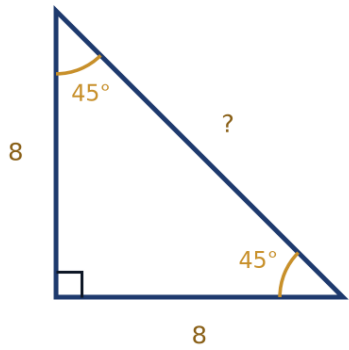


The Pythagorean theorem and special right triangles

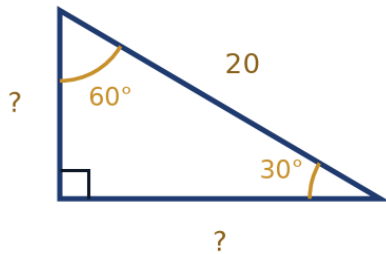
- 8 A right triangle has legs 9 and 12. Find the hypotenuse.



- 9 A 45-45-90 triangle has a leg of length 8. Find the length of the hypotenuse.

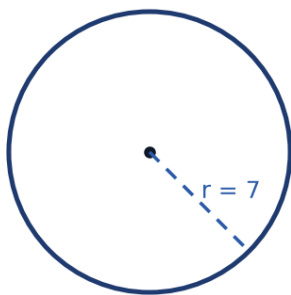


- 10 A 30-60-90 triangle has a hypotenuse of 20. Find the lengths of both legs.

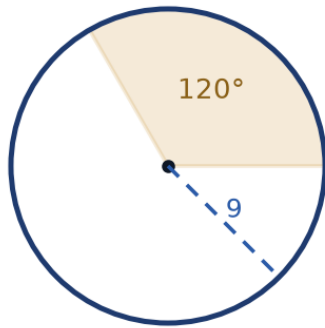


Circles: area, circumference, arcs, and sectors

- 11 Find the area and circumference of a circle with diameter 14, in terms of π .



- 12 Find the arc length of a 120° sector in a circle with radius 9, in terms of π .

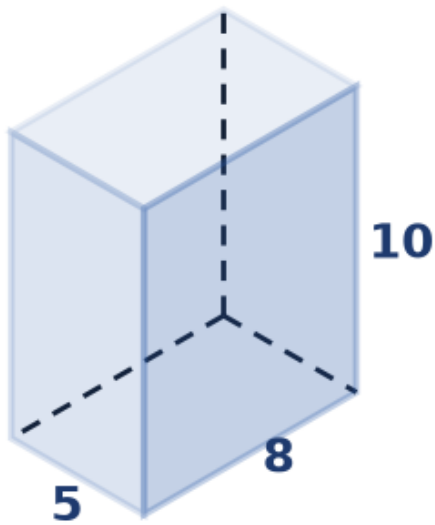


Similar and congruent figures

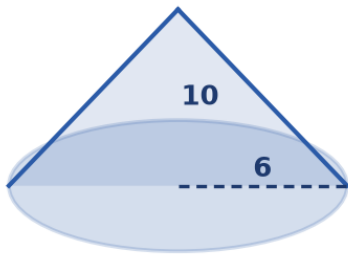
- 13 Two similar triangles have a scale factor of $\frac{3}{5}$. If the smaller triangle has area 18, find the area of the larger triangle.

Volume and surface area of 3D solids

- 14 Find the volume of a rectangular prism with dimensions 5, 8, and 10.



- 15 Find the volume of a cone with radius 6 and height 10, in terms of π .

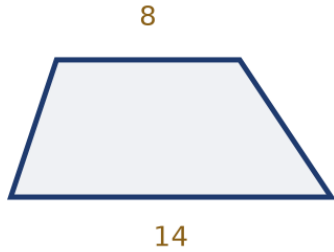


Multi-step geometry problems

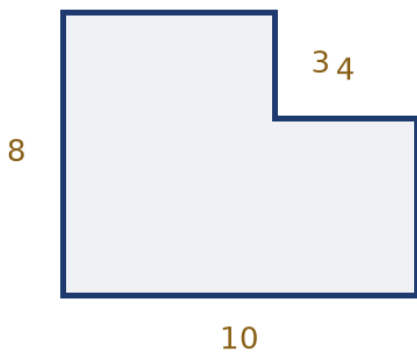
- 16 This question has two parts. A square garden has area 144 square feet. (a) Find the side length and perimeter of the garden. (b) A circular fountain with radius 3 feet is placed inside the garden. Find the remaining garden area (not covered by the fountain), in terms of π .

Trapezoids and composite polygon areas

- 17 Find the area of a trapezoid with parallel sides 8 and 14, and height 6.

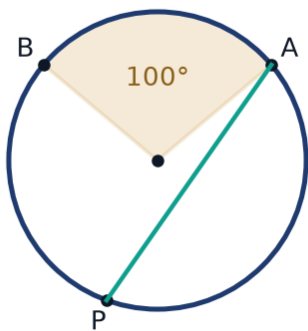


- 18 An L-shaped figure is formed by a 10-by-8 rectangle with a 4-by-3 rectangle removed from one corner. Find the remaining area.



Chords, tangents, and inscribed angles in circles

- 19 An inscribed angle subtends an arc of 100° . Find the measure of the inscribed angle.



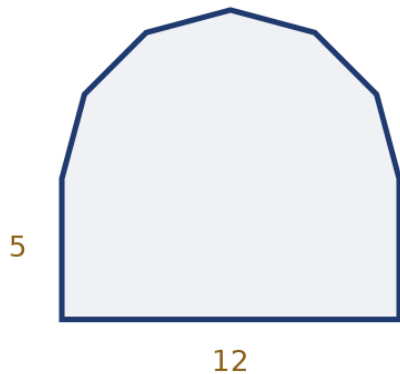
- 20 A tangent line touches a circle at point P . Explain the relationship between the tangent line and the radius drawn to point P .

Congruence and triangle proofs

- 21 State the triangle congruence criterion (SSS, SAS, ASA, AAS) that applies when two triangles share two equal angles and the included side.

A multi-part composite geometry problem

- 22 This question has two parts. A semicircular window has a diameter of 12 feet and sits atop a rectangular window of width 12 feet and height 5 feet. (a) Find the area of the semicircular portion, in terms of π . (b) Find the total area of the combined window shape.



Multiple choice — easy

- 23 Two angles are complementary, and they differ by 52° . Find the larger angle.
- a. 19°
 - b. 66°
 - c. 76°
 - d. 71°

- 24 Two angles are complementary, and they differ by 14° . Find the larger angle.
- a. 52°
 - b. 38°
 - c. 47°
 - d. 57°
- 25 Two angles are complementary, and they differ by 8° . Find the larger angle.
- a. 49°
 - b. 44°
 - c. 41°
 - d. 54°
- 26 Two angles are supplementary, and they differ by 4° . Find the larger angle.
- a. 87°
 - b. 88°
 - c. 97°
 - d. 92°
- 27 Two angles are complementary, and they differ by 48° . Find the larger angle.
- a. 74°
 - b. 64°
 - c. 21°
 - d. 69°

- 28 Two angles are supplementary, and they differ by 14° . Find the larger angle.
- a. 92°
 - b. 102°
 - c. 83°
 - d. 97°
- 29 Two angles are supplementary, and they differ by 46° . Find the larger angle.
- a. 67°
 - b. 108°
 - c. 113°
 - d. 118°
- 30 Two angles are complementary, and they differ by 32° . Find the larger angle.
- a. 61°
 - b. 56°
 - c. 29°
 - d. 66°
- 31 Two angles are supplementary, and they differ by 30° . Find the larger angle.
- a. 105°
 - b. 100°
 - c. 75°
 - d. 110°

- 32 Two angles are complementary, and they differ by 34° . Find the larger angle.
- a. 62°
 - b. 67°
 - c. 57°
 - d. 28°
- 33 Two angles are complementary, and they differ by 22° . Find the larger angle.
- a. 34°
 - b. 61°
 - c. 51°
 - d. 56°
- 34 Two angles are supplementary, and they differ by 130° . Find the larger angle.
- a. 155°
 - b. 25°
 - c. 160°
 - d. 150°
- 35 Two angles are complementary, and they differ by 76° . Find the larger angle.
- a. 88°
 - b. 7°
 - c. 78°
 - d. 83°

- 36 Two angles are supplementary, and they differ by 170° . Find the larger angle.
- a. 175°
 - b. 170°
 - c. 5°
 - d. 180°
- 37 Two angles are complementary, and they differ by 24° . Find the larger angle.
- a. 57°
 - b. 52°
 - c. 33°
 - d. 62°
-

Multiple choice — medium

- 38 A right triangle has legs 12 and 16. Find the length of the hypotenuse.
- a. 28
 - b. 21
 - c. 19
 - d. 20
- 39 A right triangle has legs 16 and 30. Find the length of the hypotenuse.
- a. 35
 - b. 34
 - c. 33
 - d. 46

- 40 A right triangle has legs 27 and 36. Find the length of the hypotenuse.
- a. 63
 - b. 44
 - c. 46
 - d. 45
- 41 A right triangle has legs 6 and 8. Find the length of the hypotenuse.
- a. 9
 - b. 11
 - c. 14
 - d. 10
- 42 A right triangle has legs 40 and 96. Find the length of the hypotenuse.
- a. 105
 - b. 136
 - c. 104
 - d. 103
- 43 A right triangle has legs 9 and 40. Find the length of the hypotenuse.
- a. 41
 - b. 42
 - c. 40
 - d. 49

- 44 A right triangle has legs 24 and 32. Find the length of the hypotenuse.
- a. 40
 - b. 39
 - c. 41
 - d. 56
- 45 A right triangle has legs 14 and 48. Find the length of the hypotenuse.
- a. 50
 - b. 62
 - c. 49
 - d. 51
- 46 A right triangle has legs 15 and 36. Find the length of the hypotenuse.
- a. 38
 - b. 40
 - c. 51
 - d. 39
- 47 A right triangle has legs 10 and 24. Find the length of the hypotenuse.
- a. 27
 - b. 26
 - c. 25
 - d. 34

- 48 A right triangle has legs 7 and 24. Find the length of the hypotenuse.
- a. 31
 - b. 24
 - c. 25
 - d. 26
- 49 A right triangle has legs 18 and 24. Find the length of the hypotenuse.
- a. 42
 - b. 31
 - c. 30
 - d. 29
- 50 A right triangle has legs 36 and 48. Find the length of the hypotenuse.
- a. 84
 - b. 60
 - c. 59
 - d. 61
- 51 A right triangle has legs 20 and 48. Find the length of the hypotenuse.
- a. 52
 - b. 68
 - c. 53
 - d. 51

52 A right triangle has legs 80 and 84. Find the length of the hypotenuse.

- a. 115
- b. 117
- c. 116
- d. 164

Multiple choice — hard

53 A circle has radius 11. Find the arc length of a 270° sector, in terms of π .

- a. $\frac{33}{2}\pi$
- b. $\frac{35}{2}\pi$
- c. $\frac{31}{2}\pi$
- d. 33π

54 A circle has radius 9. Find the arc length of a 120° sector, in terms of π .

- a. 5π
- b. 6π
- c. 12π
- d. 7π

55 A circle has radius 2. Find the sector area of a 180° sector, in terms of π .

- a. 2π
- b. 3π
- c. 4π
- d. 1π

- 56 A circle has radius 12. Find the sector area of a 180° sector, in terms of π .
- a. 144π
 - b. 71π
 - c. 73π
 - d. 72π
- 57 A circle has radius 12. Find the arc length of a 30° sector, in terms of π .
- a. 1π
 - b. 3π
 - c. 2π
 - d. 4π
- 58 Find the volume of a rectangular prism with dimensions 10, 9, and 7.
- a. 446
 - b. 620
 - c. 640
 - d. 630
- 59 Find the volume of a cylinder with radius 5 and height 14, in terms of π .
- a. 350π
 - b. 360π
 - c. 140π
 - d. 340π

- 60 Find the volume of a cylinder with radius 2 and height 15, in terms of π .
- a. 50π
 - b. 70π
 - c. 60π
 - d. 56π
- 61 Find the volume of a cone with radius 5 and height 10, in terms of π .
- a. $\frac{244}{3}\pi$
 - b. $\frac{250}{3}\pi$
 - c. $\frac{256}{3}\pi$
 - d. 250π
- 62 Find the volume of a cone with radius 8 and height 10, in terms of π .
- a. $\frac{634}{3}\pi$
 - b. 640π
 - c. $\frac{646}{3}\pi$
 - d. $\frac{640}{3}\pi$
- 63 Two similar triangles have a scale factor of $\frac{2}{5}$. If the smaller triangle has area 16, find the area of the larger triangle.
- a. 105
 - b. 40
 - c. 95
 - d. 100

- 64 Two similar triangles have a scale factor of $\frac{1}{3}$. If the smaller triangle has area 1, find the area of the larger triangle.
- a. 9
 - b. 3
 - c. 4
 - d. 14
- 65 Two similar triangles have a scale factor of $\frac{1}{2}$. If the smaller triangle has area 1, find the area of the larger triangle.
- a. 1
 - b. 9
 - c. 2
 - d. 4
- 66 Two similar triangles have a scale factor of $\frac{3}{5}$. If the smaller triangle has area 18, find the area of the larger triangle.
- a. 50
 - b. 30
 - c. 45
 - d. 55
- 67 Two similar triangles have a scale factor of $\frac{1}{2}$. If the smaller triangle has area 2, find the area of the larger triangle.
- a. 3
 - b. 13
 - c. 4
 - d. 8